

From probabilities to actions

1 Two costs make one line

If a false alarm costs C_{fa} and a miss costs C_{miss} , act whenever P is above their ratio.

act when $P > C_{fa} / (C_{fa} + C_{miss})$

umbrella 0.2 · blocking 0.048 · auto-close 0.0016

2 0.5 is a special case

It only makes sense when both mistakes cost the same. Here 0.5 cost 2.3× more than the formula's line.

3 The formula needs honest P

On the calibrated model, theory and practice agree.
On the rebalanced one, the same line cost 62% more.



4 Precision and recall trade

Lower the line: catch more, bother more. The costs tell you where to stop.

5 "I don't know" is an action

Decide only the confident cases and the error rate drops: 6.1% deciding all, 2.2% deciding the surest 80%.

6 Three doors

Act, review, escalate: machines at the easy ends, people in the middle. Chapter 21 runs it for real.

ACT

REVIEW

ESCALATE